DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

23CSD415 PROJECT PHASE I — Project Diary

ISL-LENS: Real-Time Indian Sign Language Recognition And Interactive AI Console | 2023–2027 Batch | July 2026 – Nov 2026

Name: AEVLIN PRINCE

Roll No.: 13

Reg. No.: AJC23CS014

Guide: Dr.Indu John, Asst. Professor, Dept. of CSE

Project Commencement: 14-07-2026

Project Title: ISL-LENS: Real-Time Indian Sign Language Recognition And Interactive AI Console

Official Evaluation Schedule (for reference — from the 23CSD415 Project Phase I circular)

Sr.	Date / Period	Official Milestone (23CSD415 Project Phase I)	Marks
1	6.7.2026 – 7.7.2026	Discussion with guide and finalizing the topic. Daily/weekly activity diary maintained.	10
2	8.7.2026 – 14.7.2026	Project Proposal presentation. Abstract + base paper (SCI/Scopus, 2025/26) submitted to shared folder.	—
3	3.8.2026 – 14.8.2026	Interim Review 1 — literature survey, requirement analysis, feasibility study, system design, 20% implementation.	20
4	14.9.2026 – 25.9.2026	Interim Review 2 — 50% implementation, module demonstration.	20
5	26.10.2026 – 28.10.2026	Final Review — 80% implementation, 10-min presentation + 10-min demo. Report submitted 26.10.2026 (10) + Coordinator (10).	30+10+10

Development Timeline

Week / Month	Development Work to be Done & Progress Status	Remarks	Signature of Project guide with Date
Week 1 02–06 Jul 2026	Project topic discussed with guide Dr. Indu John and finalized on 06 July 2026. Defined project objectives, technical scope, baseline feasibility, and target Deaf and Hard-of-Hearing (D/HOH) end-user requirements.		
Week 2 13–17 Jul 2026	Project proposal presentation prepared and delivered on 14 July 2026. Presented hardware-free webcam architecture, isolated/continuous ISL targets, literature benchmark comparison, and project milestone plan.		
Week 3 20–24 Jul 2026	Conducted comprehensive literature survey on vision-based sign recognition, base paper analysis (SCI/Scopus 2025/26), MediaPipe Holistic spatial coordinate normalization, and open-source ISL datasets.		
Week 4 27–31 Jul 2026	Drafted System Requirements Specification (SRS) and feasibility study. Finalized client-side browser execution pipeline, 30 FPS camera processing constraints, and zero-dependency vanilla JS architecture.		
Week 5 10–14 Aug 2026 **Interim Review 1 (20 Marks)**	Demonstrated system architecture, literature survey, SRS, and initial landmark extraction prototype during Interim Review 1 on 13 August 2026. Approximately 20% implementation verified and approved.		

Week / Month	Development Work to be Done & Progress Status	Remarks	Signature of Project guide with Date
Week 6 17–21 Aug 2026	Organized and preprocessed ISL video datasets (INCLUDE, CISLR, ISL500) across 40 vocabulary classes. Built automated keypoint extraction pipeline using MediaPipe Holistic (pose, face, hands).		
Week 7 24–28 Aug 2026	Implemented 2D landmark spatial coordinate normalization, upper-body/shoulder anchoring, and feature cache generation. Trained baseline multinomial logistic regression classifier on landmark vectors.		
Week 8 31 Aug–04 Sep 2026	Tuned multi-class sign recognition model across 40 classes. Implemented calibrated confidence thresholding (0.70) and dynamic open-set UNKNOWN rejection filter to eliminate background gesture noise.		
Week 9 07–11 Sep 2026	Developed web-native frontend interface using HTML5, CSS3, and JavaScript. Integrated live browser webcam feed, real-time landmark canvas overlay, and client-side inference pipeline.		
Week 10 14–18 Sep 2026	Constructed continuous sentence builder module, Subject-Object-Verb (SOV) to Subject-Verb-Object (SVO) grammar reordering rules, and Web Speech API speech synthesis integration.		
Week 11 21–25 Sep 2026 **Interim Review 2 (20 Marks)**	Demonstrated live webcam sign recognition, real-time sentence construction, and text-to-speech synthesis during Interim Review 2 on 24 September 2026. 50% project implementation verified.		
Week 12 28 Sep–02 Oct 2026	Developed Interactive AI Retraining Studio. Enabled users to capture custom webcam gesture exemplars, compute 828-dimensional feature vectors, and update classification weights via validation gate.		
Week 13 05–09 Oct 2026	Built Sign Practice Studio featuring authentic dataset reference video player, real-time circular HUD accuracy gauge, and auditory feedback cues. Conducted review on 08 October 2026.		
Week 14 12–16 Oct 2026	Conducted quantitative system profiling: achieved 18.4 ms recognition latency (54.3 FPS throughput), 96.88% cross-signer test accuracy, and validated lighting and background invariance.		
Week 15 19–23 Oct 2026	Prepared comprehensive project documentation, draft B.Tech report, IJERA journal research paper manuscript, and presentation slides. Rehearsed final evaluation demonstration on 22 October 2026.		
Week 16 26–30 Oct 2026 **Final Review (30+10+10 Marks)**	Delivered final 10-minute presentation and live 10-minute system demonstration before the evaluation committee on 28 October 2026. Successfully defended 80% implementation and submitted report.		
Week 17 02–06 Nov 2026	Incorporated evaluation committee feedback, finalized project report and camera-ready journal paper, and formulated Phase II (Semester 8) deep learning and sentence expansion roadmap.		

Signature of Student **Signature of Project guide**